

11.3.2 Conical Spiral

The shape of a nonplanar spiral can be described by defining the derivative of $f(\theta)$ to be

$$\frac{df}{d\theta} = f'(\theta) = A\delta(\beta - \theta) \quad (11-19)$$

in which β is allowed to take any value in the range $0 \leq \beta \leq \pi$. For a given value of β , (11-19) in conjunction with (11-8) describes a spiral wrapped on a conical surface. The edges of one conical spiral surface are defined by

$$r_2 = r'_2 e^{(a \sin \theta_0) \phi} = r'_2 e^{b \phi} \quad (11-20a)$$

$$r_3 = r'_3 e^{a \sin \theta_0 \phi} = r'_2 e^{a \sin \theta_0 (\phi - \delta)} \quad (11-20b)$$

where

$$r'_3 = r'_2 e^{-(a \sin \theta_0) \delta} \quad (11-20c)$$

and θ_0 is half of the total included cone angle. Larger values of θ_0 in $0 \leq \theta \leq \pi/2$ represent less tightly wound spirals. These equations correspond to (11-15a)–(11-15c) for the planar surface. The second arm of a balanced system can be defined by shifting each of (11-20a)–(11-20c) by 180° , as was done for the planar surface by (11-17)–(11-18a). A conical spiral metal strip antenna of elliptical polarization is shown in Figure 11.4.

The conducting conical spiral surface can be constructed conveniently by forming, using printed-circuit techniques, the conical arms on the dielectric cone which is also used as a support. The feed cable can be bonded to the metal arms which are wrapped around the cone. Symmetry can be preserved by observing the same precautions, like the use of a dummy cable, as was done for the planar surface.

A distinct difference between the planar and conical spirals is that the latter provides unidirectional radiation (single lobe) toward the apex of the cone with the maximum along the axis. Circular polarization and relatively constant impedances are preserved over large bandwidths. Smoother patterns have been observed for unidirectional designs. Conical spirals can be used in conjunction with a ground plane, with a reduction in bandwidth when they are flush mounted on the plane.

11.4 LOG-PERIODIC ANTENNAS

Another type of an antenna configuration, which closely parallels the frequency independent concept, is the log-periodic structure introduced by DuHamel and Isbell [4]. Because the entire shape of it cannot be solely specified by angles, it is not truly frequency independent.

11.4.1 Planar and Wire Surfaces

A planar log-periodic structure is shown in Figure 11.5(a). It consists of a metal strip whose edges are specified by the angle $\alpha/2$. However, in order to specify the length from the origin to any point on the structure, a distance characteristic must be included.



Figure 11.4 Conical spiral metal strip antenna. (SOURCE: *Antennas, Antenna Masts and Mounting Adaptors*, American Electronic Laboratories, Inc., Lansdale, Pa., Catalog 7.5M-7-79. Courtesy of American Electronic Laboratories, Inc., Montgomeryville, PA 18936 USA).

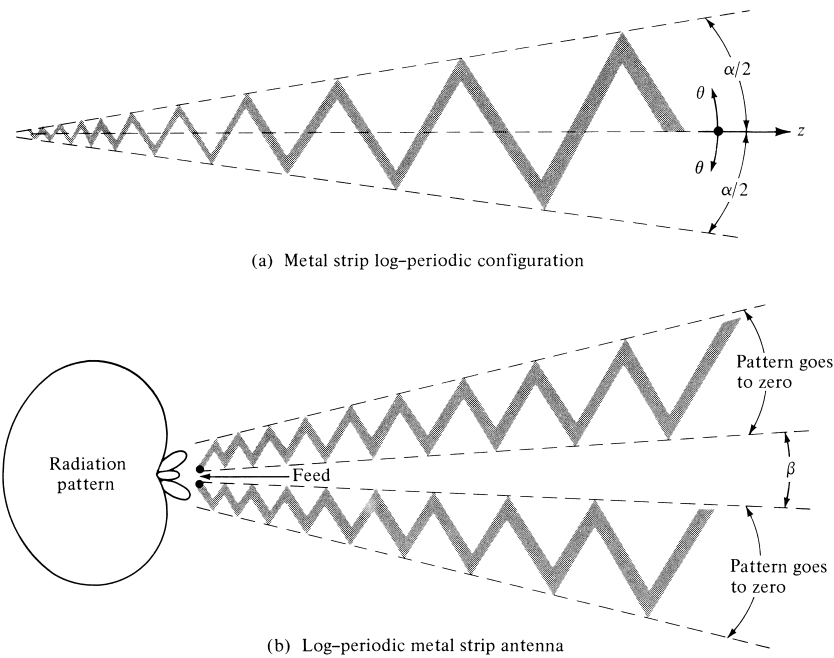


Figure 11.5 Typical metal strip log-periodic configuration and antenna structure.

In spherical coordinates (r, θ, ϕ) the shape of the structure can be written as

$$\theta = \text{periodic function of } [b \ln(r)] \quad (11-21)$$

An example of it would be

$$\theta = \theta_0 \sin \left[b \ln \left(\frac{r}{r_0} \right) \right] \quad (11-22)$$

It is evident from (11-22) that the values of θ are repeated whenever the logarithm of the radial frequency $\ln(\omega) = \ln(2\pi f)$ differs by $2\pi/b$. The performance of the system is then periodic as a function of the logarithm of the frequency; thus the name *logarithmic-periodic or log-periodic*.

A typical log-periodic antenna configuration is shown in Figure 11.5(b). It consists of two coplanar arms of the Figure 11.5(a) geometry. The pattern is unidirectional toward the apex of the cone formed by the two arms, and it is linearly polarized. Although the patterns of this and other log-periodic structures are not completely frequency independent, the amplitude variations of certain designs are very slight. Thus practically they are frequency independent.

Log-periodic wire antennas were introduced by DuHamel [4]. While investigating the current distribution on log-periodic surface structures of the form shown in Figure 11.6(a), he discovered that the fields on the conductors attenuated very sharply with distance. This suggested that perhaps there was a strong current concentration at or near the edges of the conductors. Thus removing part of the inner surface to form a wire antenna as shown in Figure 11.6(b) should not seriously degrade the performance of the antenna. To verify this, a wire antenna, with geometrical shape identical to the pattern formed by the edges of the conducting surface, was built and it was investigated

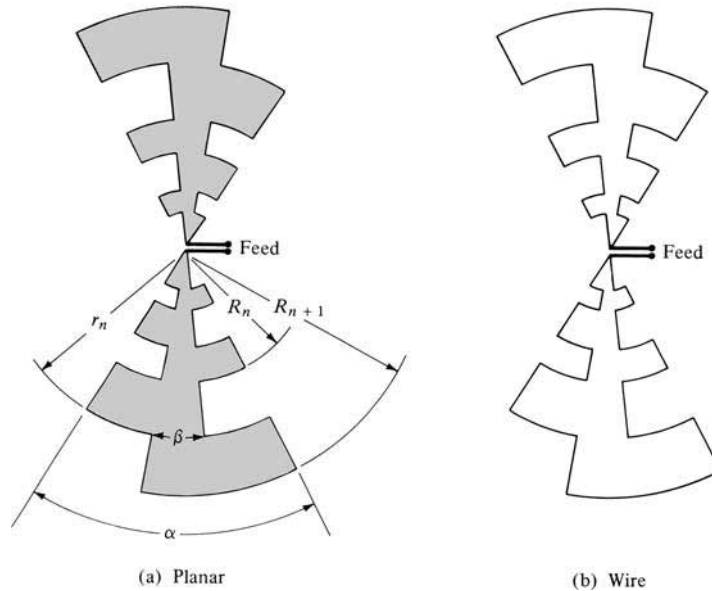


Figure 11.6 Planar and wire logarithmically periodic antennas.

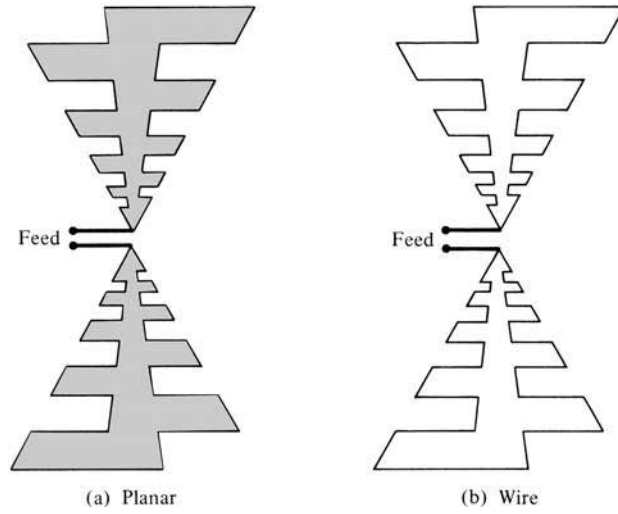


Figure 11.7 Planar and wire trapezoidal toothed log-periodic antennas.

experimentally. As predicted, it was found that the performance of this antenna was almost identical to that of Figure 11.6(a); thus the discovery of a much simpler, lighter in weight, cheaper, and less wind resistant antenna. Nonplanar geometries in the form of a V, formed by bending one arm relative to the other, are also widely used.

If the wires or the edges of the plates are linear (instead of curved), the geometries of Figure 11.6 reduce, respectively, to the trapezoidal tooth log-periodic structures of Figure 11.7. These simplifications result in more convenient fabrication geometries with basically no loss in operational performance. There are numerous other bizarre but practical configurations of log-periodic structures, including log-periodic arrays.

If the geometries of Figure 11.6 use uniform periodic teeth, we define the geometric ratio of the log-periodic structure by

$$\tau = \frac{R_n}{R_{n+1}} \quad (11-23)$$

and the width of the antenna slot by

$$\chi = \frac{r_n}{R_{n+1}} \quad (11-24)$$

The geometric ratio τ of (11-23) defines the period of operation. For example, if two frequencies f_1 and f_2 are one period apart, they are related to the geometric ratio τ by

$$\tau = \frac{f_1}{f_2}, \quad f_2 > f_1 \quad (11-25)$$

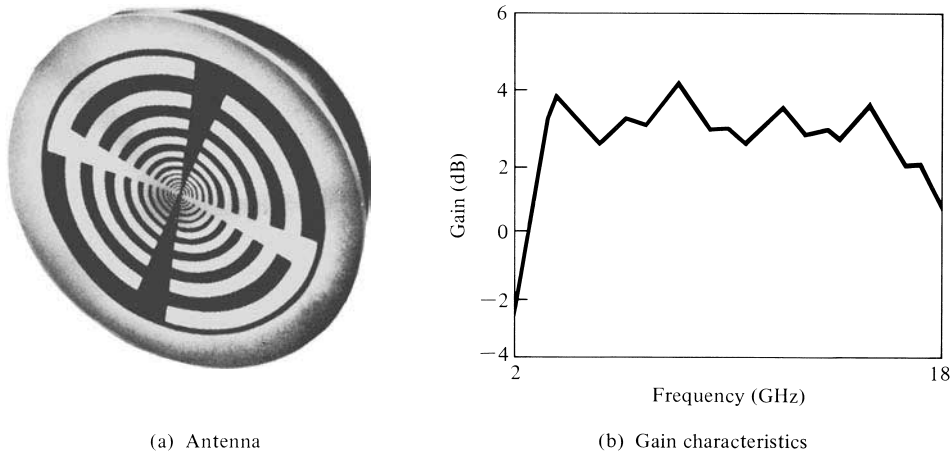


Figure 11.8 Linearly polarized flush-mounted cavity-backed log-periodic slot antenna and typical gain characteristics. (SOURCE: *Antennas, Antenna Masts and Mounting Adaptors*, American Electronic Laboratories, Inc., Lansdale, Pa., Catalog 7.5M-7-79. Courtesy of American Electronic Laboratories, Inc., Montgomeryville, PA 18936 USA).

Extensive studies on the performance of the antenna of Figure 11.6(b) as a function of α , β , τ , and χ , have been performed [9]. In general, these structures performed almost as well as the planar and conical structures. The only major difference is that the log-periodic configurations are linearly polarized instead of circular.

A commercial lightweight, cavity-backed, linearly polarized, flush-mounted log-periodic slot antenna and its associated gain characteristics are shown in Figures 11.8(a) and (b). Typical electrical characteristics are: VSWR—2:1; E -plane beamwidth— 70° ; H -plane beamwidth— 70° . The maximum diameter of the cavity is about 2.4 in. (6.1 cm), the depth is 1.75 in. (4.445 cm), and the weight is near 5 oz (0.14 kg).

11.4.2 Dipole Array

To the layman, the most recognized log-periodic antenna structure is the configuration introduced by Isbell [5] which is shown in Figure 11.9(a). It consists of a sequence of side-by-side parallel linear dipoles forming a coplanar array. Although this antenna has slightly smaller directivities than the Yagi–Uda array (7–12 dB), they are achievable and maintained over much wider bandwidths. There are, however, major differences between them.

While the geometrical dimensions of the Yagi–Uda array elements do not follow any set pattern, the lengths (l_n 's), spacings (R_n 's), diameters (d_n 's), and even gap spacings at dipole centers (s_n 's) of the log-periodic array increase logarithmically as defined by the inverse of the geometric ratio τ . That is,

$$\frac{1}{\tau} = \frac{l_2}{l_1} = \frac{l_{n+1}}{l_n} = \frac{R_2}{R_1} = \frac{R_{n+1}}{R_n} = \frac{d_2}{d_1} = \frac{d_{n+1}}{d_n} = \frac{s_2}{s_1} = \frac{s_{n+1}}{s_n} \quad (11-26)$$

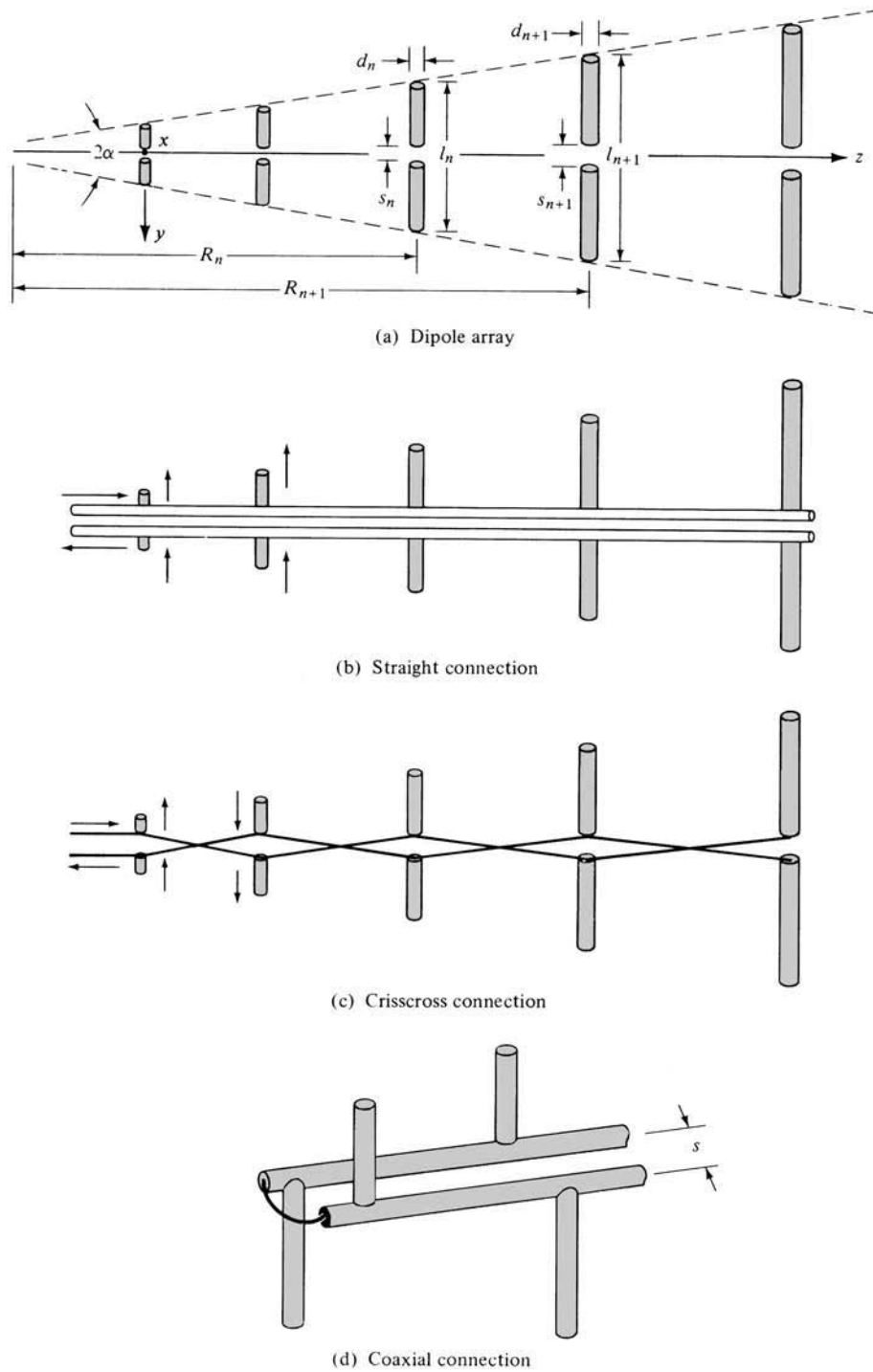


Figure 11.9 Log-periodic dipole array and associated connections.

Another parameter that is usually associated with a log-periodic dipole array is the spacing factor σ defined by

$$\sigma = \frac{R_{n+1} - R_n}{2l_{n+1}} \quad (11-26a)$$

Straight lines through the dipole ends meet to form an angle 2α which is a characteristic of frequency independent structures.

Because it is usually very difficult to obtain wires or tubing of many different diameters and to maintain tolerances of very small gap spacings, constant dimensions in these can be used. These relatively minor factors will not sufficiently degrade the overall performance.

While only one element of the Yagi–Uda array is directly energized by the feed line, while the others operate in a parasitic mode, all the elements of the log-periodic array are connected. There are two basic methods, as shown in Figures 11.9(b) and 11.9(c), which could be used to connect and feed the elements of a log-periodic dipole array. In both cases the antenna is fed at the small end of the structure.

The currents in the elements of Figure 11.9(b) have the same phase relationship as the terminal phases. If in addition the elements are closely spaced, the phase progression of the currents is to the right. This produces an end-fire beam in the direction of the longer elements and interference effects to the pattern result.

It was recognized that by mechanically crisscrossing or transposing the feed between adjacent elements, as shown in Figure 11.9(c), a 180° phase is added to the terminal of each element. Since the phase between the adjacent closely spaced short elements is almost in opposition, very little energy is radiated by them and their interference effects are negligible. However, at the same time, the longer and larger spaced elements radiate. The mechanical phase reversal between these elements produces a phase progression so that the energy is beamed end fire in the direction of the shorter elements. The most active elements for this feed arrangement are those that are near resonant with a combined radiation pattern toward the vertex of the array.

The feed arrangement of Figure 11.9(c) is convenient provided the input feed line is a balanced line like the two-conductor transmission line. Using a coaxial cable as a feed line, a practical method to achieve the 180° phase reversal between adjacent elements is shown in Figure 11.9(d). This feed arrangement provides a built-in broadband balun resulting in a balanced overall system. The elements and the feeder line of this array are usually made of piping. The coaxial cable is brought to the feed through the hollow part of one of the feeder-line pipes. While the outside conductor of the coax is connected to that conductor at the feed, its inner conductor is extended and it is connected to the other pipe of the feeder line.

If the geometrical pattern of the log-periodic array, as defined by (11-26), is to be maintained to achieve a truly log-periodic configuration, an infinite structure would result. However, to be useful as a practical broadband radiator, the structure is truncated at both ends. This limits the frequency of operation to a given bandwidth.

The cutoff frequencies of the truncated structure can be determined by the electrical lengths of the longest and shortest elements of the structure. The lower cutoff frequency occurs approximately when the longest element is $\lambda/2$; however, the high cutoff frequency occurs when the shortest element is nearly $\lambda/2$ only when the active

region is very narrow. Usually it extends beyond that element. The active region of the log-periodic dipole array is near the elements whose lengths are nearly or slightly smaller than $\lambda/2$. The role of active elements is passed from the longer to the shorter elements as the frequency increases. Also the energy from the shorter active elements traveling toward the longer inactive elements decreases very rapidly so that a negligible amount is reflected from the truncated end. The movement of the active region of the antenna, and its associated phase center, is an undesirable characteristic in the design of feeds for reflector antennas (see Chapter 15). For this reason, log-periodic arrays are not widely used as feeds for reflectors.

The decrease of energy toward the longer inactive elements is demonstrated in Figure 11.10(a). The curves represent typical computed and measured transmission-line voltages (amplitude and phase) on a log-periodic dipole array [10] as a function of distance from its apex. These are feeder-line voltages at the base of the elements of an array with $\tau = 0.95$, $\sigma = 0.0564$, $N = 13$, and $l_n/d_n = 177$. The frequency of operation is such that element No. 10 is $\lambda/2$. The amplitude voltage is nearly constant from the first (the feed) to the eighth element while the corresponding phase is uniformly progressive. Very rapid decreases in amplitude and nonlinear phase variations are noted beyond the eighth element.

The region of constant voltage along the structure is referred to as the *transmission region*, because it resembles that of a matched transmission line. Along the structure, there is about 150° phase change for every $\lambda/4$ free-space length of transmission line. This indicates that the phase velocity of the wave traveling along the structure is $v_p = 0.6v_0$, where v_0 is the free-space velocity. The smaller velocity results from the shunt capacitive loading of the line by the smaller elements. The loading is almost constant per unit length because there are larger spacings between the longer elements.

The corresponding current distribution is shown in Figure 11.10(b). It is noted that the rapid decrease in voltage is associated with strong current excitation of elements 7–10 followed by a rapid decline. The region of high current excitation is designated as the *active region*, and it encompasses 4 to 5 elements for this design. The voltage and current excitations of the longer elements (beyond the ninth) are relatively small, reassuring that the truncated larger end of the structure is not affecting the performance. The smaller elements, because of their length, are not excited effectively. As the frequency changes, the relative voltage and current patterns remain essentially the same, but they move toward the direction of the active region.

There is a linear increase in current phase, especially in the active region, from the shorter to the longer elements. This phase shift progression is opposite in direction to that of an unloaded line. It suggests that on the log-periodic antenna structure there is a wave that travels toward the feed forming a unidirectional end-fire pattern toward the vertex.

The radiated wave of a single log-periodic dipole array is linearly polarized, and it has horizontal polarization when the plane of the antenna is parallel to the ground. Bidirectional patterns and circular polarization can be obtained by phasing multiple log-periodic dipole arrays. For these, the overall effective phase center can be maintained at the feed.

If the input impedance of a log-periodic antenna is plotted as a function of frequency, it will be repetitive. However, if it is plotted as a function of the *logarithm* of the frequency, it will be *periodic* (not necessarily sinusoidal) with each cycle being exactly identical to the preceding one. Hence the name *log-periodic*, because the variations

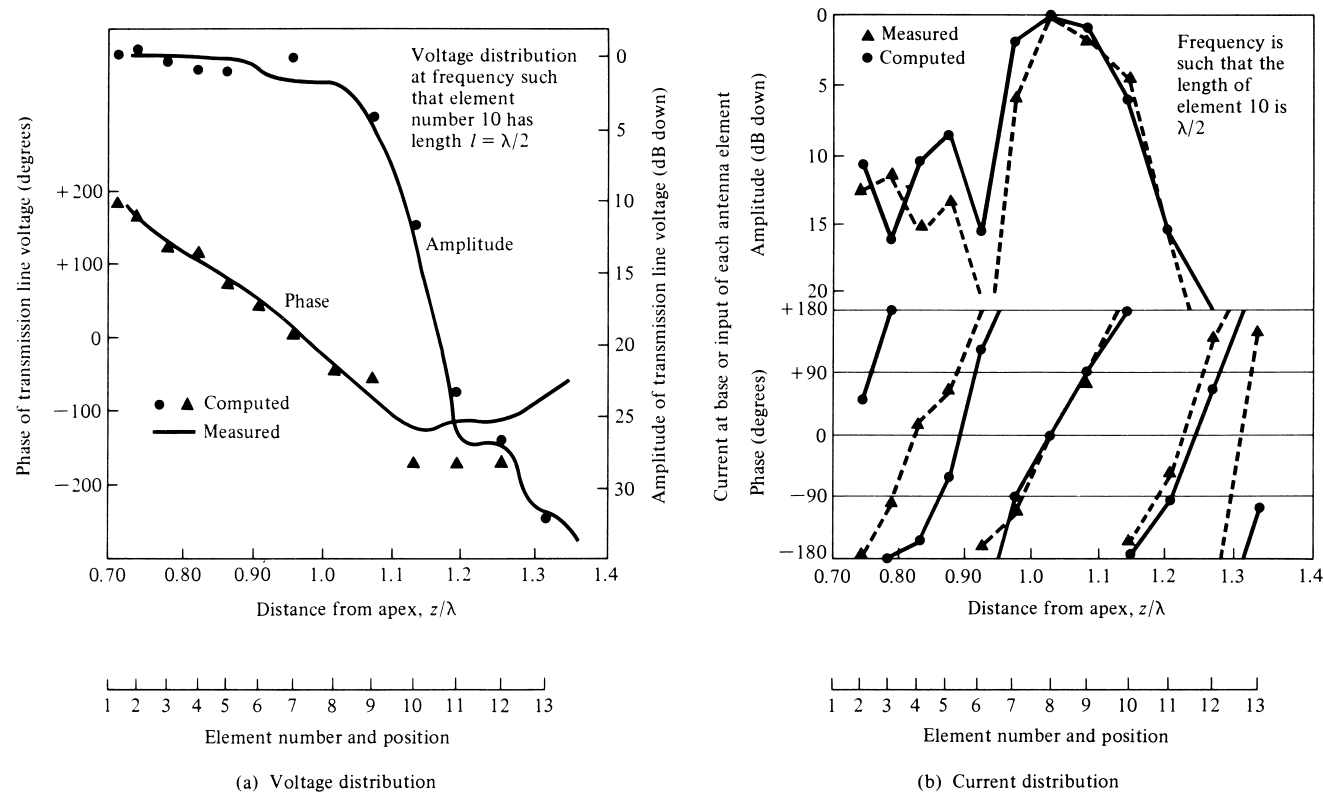


Figure 11.10 Measured and computed voltage and current distributions on a log-periodic dipole array of 13 elements with frequency such that $l_{10} = \lambda/2$. (SOURCE: R. L. Carrel, "Analysis and Design of the Log-Periodic Dipole Antenna," Ph.D. Dissertation, Elec. Eng. Dept., University of Illinois, 1961, University Microfilms, Inc., Ann Arbor, Michigan).

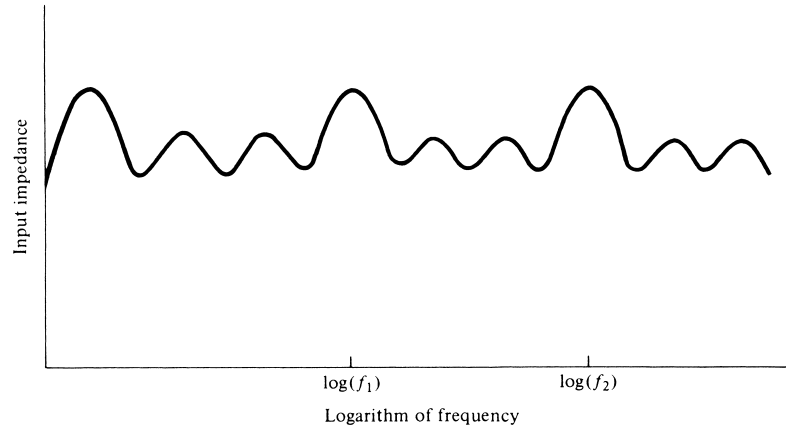


Figure 11.11 Typical input impedance variation of a log-periodic antenna as a function of the logarithm of the frequency.

are *periodic* with respect to the *logarithm* of the frequency. A typical variation of the impedance as a function of frequency is shown in Figure 11.11. Other parameters that undergo similar variations are the pattern, directivity, beamwidth, and side lobe level.

The periodicity of the structure does not ensure broadband operation. However, if the variations of the impedance, pattern, directivity, and so forth within one cycle are made sufficiently small and acceptable for the corresponding bandwidth of the cycle, broadband characteristics are ensured within acceptable limits of variation. The total bandwidth is determined by the number of repetitive cycles for the given truncated structure.

The relative frequency span Δ of each cycle is determined by the geometric ratio as defined by (11-25) and (11-26).^{*} Taking the logarithm of both sides in (11-25) reduces to

$$\Delta = \ln(f_2) - \ln(f_1) = \ln\left(\frac{1}{\tau}\right) \quad (11-27)$$

The variations that occur within a given cycle ($f_1 \leq f \leq f_2 = f_1/\tau$) will repeat identically at other cycles of the bandwidth defined by $f_1/\tau^{n-1} \leq f \leq f_1/\tau^n$, $n = 1, 2, 3, \dots$

Typical designs of log-periodic dipole arrays have apex half angles of $10^\circ \leq \alpha \leq 45^\circ$ and $0.95 \geq \tau \geq 0.7$. There is a relation between the values of α and τ . As α increases, the corresponding τ values decrease, and vice versa. Larger values of α or smaller values of τ result in more compact designs which require smaller number of elements separated by larger distances. In contrast, smaller values of α or larger values of τ require a larger number of elements that are closer together. For this type of a design, there are more elements in the active region which are nearly $\lambda/2$. Therefore the variations of the impedance and other characteristics as a function of frequency are smaller, because of the smoother transition between the elements, and the gains are larger.

Experimental models of log-periodic dipole arrays have been built and measurements were made [6]. The input impedances (purely resistive) and corresponding directivities

^{*}In some cases, the impedance (but not the pattern) may vary with a period which is one-half of (11-27). That is, $\Delta = \frac{1}{2} \ln(1/\tau)$.

TABLE 11.1 Input Resistances (R_{in} in ohms) and Directivities (dB above isotropic) for Log-Periodic Dipole Arrays

α	$\tau = 0.81$		$\tau = 0.89$		$\tau = 0.95$	
	$R_{in}(\text{ohms})$	$D_0(\text{dB})$	$R_{in}(\text{ohms})$	$D_0(\text{dB})$	$R_{in}(\text{ohms})$	$D_0(\text{dB})$
10	98	—	82	9.8	77.5	10.7
12.5	—	—	77	—	—	—
15	—	7.2	—	—	—	—
17.5	—	—	76	7.7	62	8.8
20	—	—	74	—	—	—
25	—	—	63	7.2	—	8.0
30	80	—	64	—	54	—
35	—	—	56	6.5	—	—
45	65	5.2	59	6.2	—	—

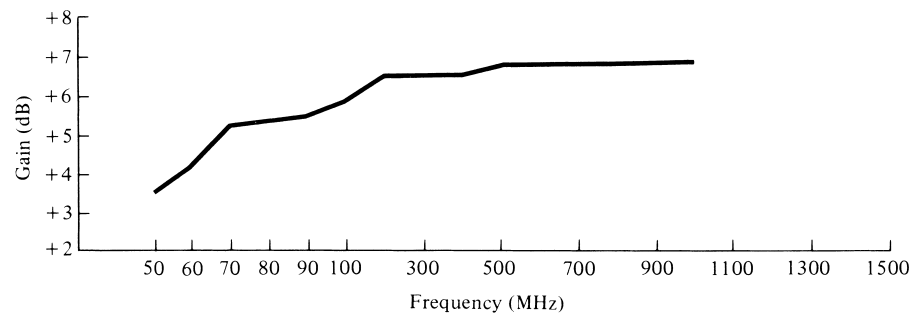
(SOURCE: D. E. Isbell, "Log Periodic Dipole Arrays," *IRE Trans. Antennas Propagat.*, Vol. AP-8, pp. 260–267, May 1960. © (1960) IEEE.)

(above isotropic) for three different designs are listed in Table 11.1. Larger directivities can be achieved by arraying multiple log-periodic dipole arrays. There are other configurations of log-periodic dipole array designs, including those with V instead of linear elements [11]. This array provides moderate bandwidths with good directivities at the higher frequencies, and it is widely used as a single TV antenna covering the entire frequency spectrum from the lowest VHF channel (54 MHz) to the highest UHF (806 MHz). Typical gain, VSWR, and E - and H -plane half-power beamwidths of commercial log-periodic dipole arrays are shown in Figures 11.12(a), (b), (c), respectively. The overall length of each of these antennas is about 105 in. (266.70 cm) while the largest element in each has an overall length of about 122 in. (309.88 cm). The weight of each antenna is about 31 lb ($\simeq 14$ kg).

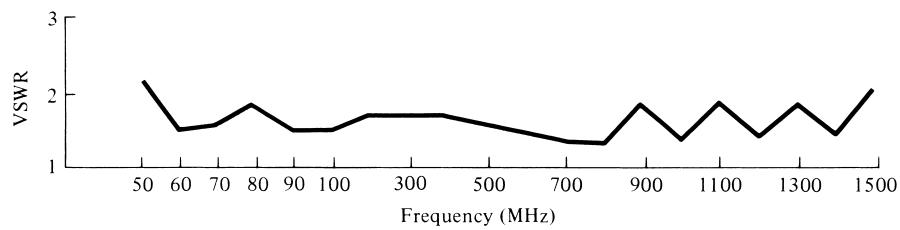
11.4.3 Design of Dipole Array

The ultimate goal of any antenna configuration is the design that meets certain specifications. Probably the most introductory, complete, and practical design procedure for a log-periodic dipole array is that by Carrel [10]. To aid in the design, he has a set of curves and nomographs. The general configuration of a log-periodic array is described in terms of the design parameters τ , α , and σ related by

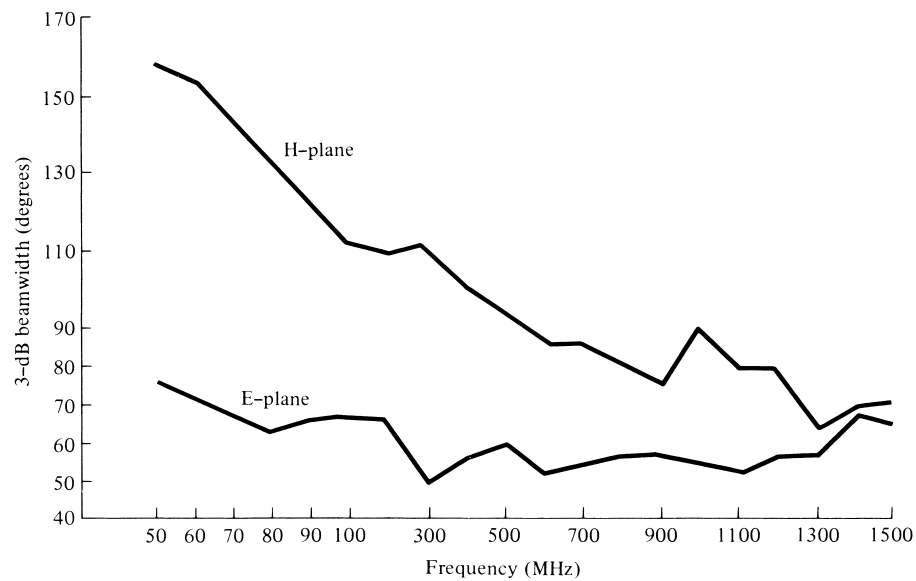
$$\alpha = \tan^{-1} \left[\frac{1 - \tau}{4\sigma} \right] \quad (11-28)$$



(a) Gain



(b) VSWR



(c) Half-power beamwidth

Figure 11.12 Typical gain, VSWR, and half-power beamwidth of commercial log-periodic dipole arrays. (SOURCE: *Antennas, Antenna Masts and Mounting Adaptors*, American Electronic Laboratories, Inc., Lansdale, Pa., Catalog 7.5M-7-79. Courtesy of American Electronic Laboratories, Inc., Montgomeryville, PA 18936 USA).

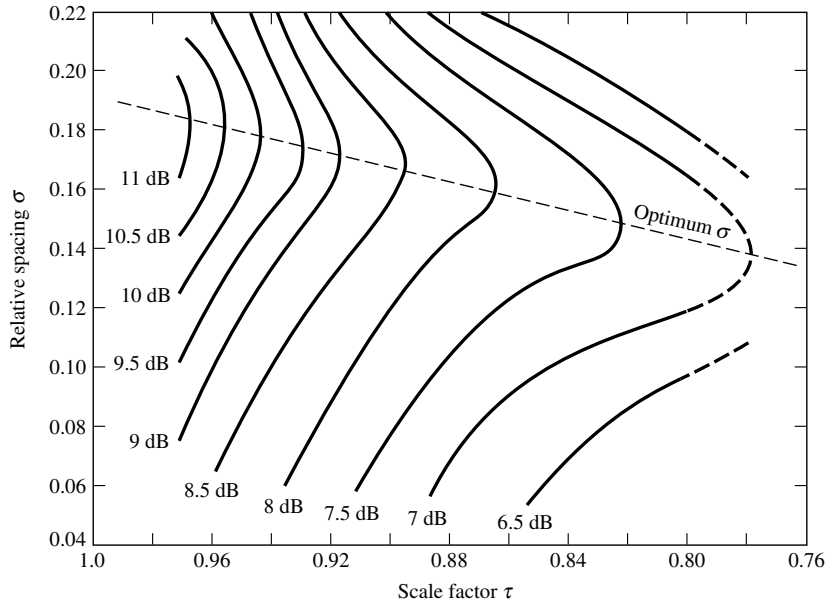


Figure 11.13 Computed contours of constant directivity versus σ and τ for log-periodic dipole arrays. (SOURCE: R. L. Carrel, "Analysis and Design of the Log-Periodic Dipole Antenna," Ph.D. Dissertation, Elec. Eng. Dept., University of Illinois, 1961, University Microfilms, Inc., Ann Arbor Michigan). *Note:* The initial curves led to designs whose directivities are 1–2 dB too high. They have been reduced by an average of 1 dB (see P. C. Butson and G. T. Thompson, "A Note on the Calculation of the Gain of Log-Periodic Dipole Antennas," *IEEE Trans. Antennas Propagat.*, AP-24, pp. 105–106, January 1976).

Once two of them are specified, the other can be found. Directivity (in dB) contour curves as a function of τ for various values of σ are shown in Figure 11.13.

The original directivity contour curves in [10] are in error because the expression for the E -plane field pattern in [10] is in error. To correct the error, the leading $\sin(\theta)$ function in front of the summation sign of equation 47 in [10] should be in the denominator and not in the numerator [i.e., replace $\sin \theta$ by $1/\sin(\theta)$] [12]. The influence of this error in the contours of Figure 11.13 is variable and leads to 1–2 dB higher directivities. However it has been suggested that, as an average, the directivity of each original contour curve be reduced by about 1 dB. This has been implemented already, and the curves in Figure 11.13 are more accurate as they now appear.

A. Design Equations

In this section a number of equations will be introduced that can be used to design a log-periodic dipole array.

While the bandwidth of the system determines the lengths of the shortest and longest elements of the structure, the width of the active region depends on the specific design. Carrel [10] has introduced a semiempirical equation to calculate the bandwidth of the active region B_{ar} related to α and τ by

$$B_{ar} = 1.1 + 7.7(1 - \tau)^2 \cot \alpha \quad (11-29)$$

In practice a slightly larger bandwidth (B_s) is usually designed than that which is required (B). The two are related by

$$B_s = BB_{ar} = B[1.1 + 7.7(1 - \tau)^2 \cot \alpha] \quad (11-30)$$

where

B_s = designed bandwidth

B = desired bandwidth

B_{ar} = active region bandwidth

The total length of the structure L , from the shortest ($l_{\min}$) to the longest ($l_{\max}$) element, is given by

$$L = \frac{\lambda_{\max}}{4} \left(1 - \frac{1}{B_s} \right) \cot \alpha \quad (11-31)$$

where

$$\lambda_{\max} = 2l_{\max} = \frac{v}{f_{\min}} \quad (11-31a)$$

From the geometry of the system, the number of elements are determined by

$$N = 1 + \frac{\ln(B_s)}{\ln(1/\tau)} \quad (11-32)$$

The center-to-center spacing s of the feeder-line conductors can be determined by specifying the required input impedance (assumed to be real), and the diameter of the dipole elements and the feeder-line conductors. To accomplish this, we first define an average characteristic impedance of the elements given by

$$Z_a = 120 \left[\ln \left(\frac{l_n}{d_n} \right) - 2.25 \right] \quad (11-33)$$

where l_n/d_n is the length-to-diameter ratio of the n th element of the array. For an ideal log-periodic design, this ratio should be the same for all the elements of the array. Practically, however, the elements are usually divided into one, two, three, or more groups with all the elements in each group having the same diameter but not the same length. The number of groups is determined by the total number of elements of the array. Usually three groups (for the small, middle, and large elements) should be sufficient.

The effective loading of the dipole elements on the input line is characterized by the graphs shown in Figure 11.14 where

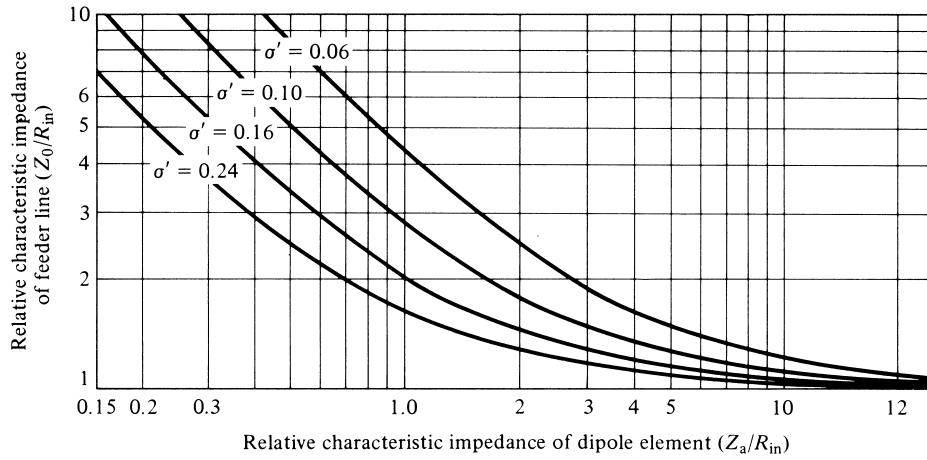


Figure 11.14 Relative characteristic impedance of a feeder line as a function of relative characteristic impedance of dipole element. (SOURCE: R. L. Carrel, "Analysis and Design of the Log-Periodic Dipole Antenna," Ph.D. Dissertation, Elec. Eng. Dept., University of Illinois, 1961, University Microfilms, Inc., Ann Arbor, Michigan).

$\sigma' = \sigma/\sqrt{\tau}$ = relative mean spacing

Z_a = average characteristic impedance of the elements

R_{in} = input impedance (real)

Z_0 = characteristic impedance of the feeder line

The center-to-center spacing s between the two rods of the feeder line, each of identical diameter d , is determined by

$$s = d \cosh \left(\frac{Z_0}{120} \right) \quad (11-34)$$

B. Design Procedure

A design procedure is outlined here, based on the equations introduced above and in the previous page, and assumes that the directivity (in dB), input impedance R_{in} (real), diameter of elements of feeder line (d), and the lower and upper frequencies ($B = f_{max}/f_{min}$) of the bandwidth are specified. It then proceeds as follows:

1. Given D_0 (dB), determine σ and τ from Figure 11.13.
2. Determine α using (11-28).
3. Determine B_{ar} using (11-29) and B_s using (11-30).
4. Find L using (11-31) and N using (11-32).
5. Determine Z_a using (11-33) and $\sigma' = \sigma/\sqrt{\tau}$.
6. Determine Z_0/R_{in} using Figure 11.14.
7. Find s using (11-34).

Example 11.1

Design a log-periodic dipole antenna, of the form shown in Figure 11.9(d), to cover all the VHF TV channels (starting with 54 MHz for channel 2 and ending with 216 MHz for channel 13. See Appendix IX.) The desired directivity is 8 dB and the input impedance is 50 ohms (ideal for a match to 50-ohm coaxial cable). The elements should be made of aluminum tubing with $\frac{3}{4}$ in. (1.9 cm) outside diameter for the largest element and the feeder line and $\frac{3}{16}$ in. (0.48 cm) for the smallest element. These diameters yield identical l/d ratios for the smallest and largest elements.

Solution:

1. From Figure 11.13, for $D_0 = 8$ dB the optimum σ is $\sigma = 0.157$ and the corresponding τ is $\tau = 0.865$.
2. Using (11-28)

$$\alpha = \tan^{-1} \left[\frac{1 - 0.865}{4(0.157)} \right] = 12.13^\circ \simeq 12^\circ$$

3. Using (11-29)

$$B_{ar} = 1.1 + 7.7(1 - 0.865)^2 \cot(12.13^\circ) = 1.753$$

and from (11-30)

$$B_s = BB_{ar} = \frac{216}{54}(1.753) = 4(1.753) = 7.01$$

4. Using (11-31a)

$$\lambda_{\max} = \frac{v}{f_{\min}} = \frac{3 \times 10^8}{54 \times 10^6} = 5.556 \text{ m (18.227 ft)}$$

From (11-31)

$$L = \frac{5.556}{4} \left(1 - \frac{1}{7.01} \right) \cot(12.13^\circ) = 5.541 \text{ m (18.178 ft)}$$

and from (11-32)

$$N = 1 + \frac{\ln(7.01)}{\ln(1/0.865)} = 14.43 \text{ (14 or 15 elements)}$$

$$5. \sigma' = \frac{\sigma}{\sqrt{\tau}} = \frac{0.157}{\sqrt{0.865}} = 0.169$$

At the lowest frequency

$$l_{\max} = \frac{\lambda_{\max}}{2} = \frac{18.227}{2} = 9.1135 \text{ ft}$$

$$\frac{l_{\max}}{d_{\max}} = \frac{9.1135(12)}{0.75} = 145.816$$

Using (11-33)

$$Z_a = 120[\ln(145.816) - 2.25] = 327.88 \text{ ohms}$$

Thus

$$\frac{Z_a}{R_{in}} = \frac{327.88}{50} = 6.558$$

6. From Figure 11.14

$$Z_0 \simeq 1.2R_{in} = 1.2(50) = 60 \text{ ohms}$$

7. Using (11-34), assuming the feeder line conductor is made of the same size tubing as the largest element of the array, the center-to-center spacing of the feeder conductors is

$$s = \frac{3}{4} \cosh\left(\frac{60}{120}\right) = 0.846 \simeq 0.85 \text{ in.}$$

which allows for a 0.1-in. separation between their conducting surfaces.

For such a high-gain antenna, this is obviously a good practical design. If a lower gain is specified and designed for, a smaller length will result.

A commercial log-periodic dipole antenna of 21 elements is shown in Figure 11.15. The antenna is designed to operate in the 100–1,100 MHz with a gain of about 6 dBi,

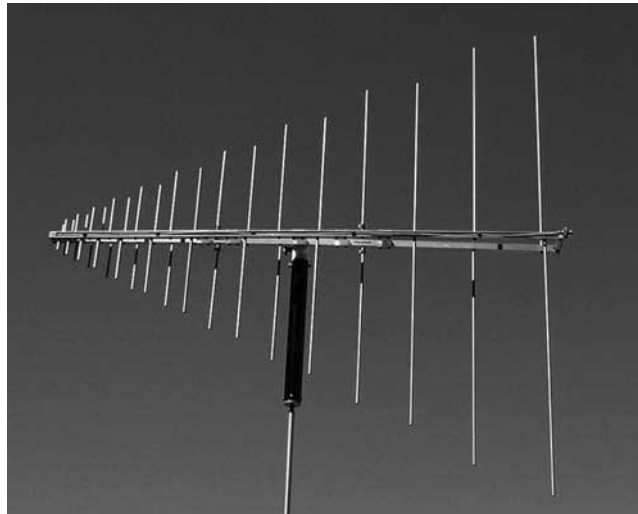


Figure 11.15 Commercial log-periodic dipole antenna of 21 elements. (Courtesy: Antenna Research Associates, Inc., Beltsville, MD).

a VSWR (to a 50-ohm line) of 2:1 maximum, a front-to-back ratio of about 20 dB, and with typical half-power beamwidths of about: E -plane (75°) and H -plane (120°).

C. Design and Analysis Computer Program

A computer program Log-Periodic Dipole Array (*log_perd*) has been developed in FORTRAN and translated in MATLAB, based primarily on the design equations of (11-28)–(11-34), and Figures 11.13 and 11.14, to design a log-periodic dipole array whose geometry is shown in Figure 11.9(a). Although most of the program is based on the same design equations as outlined in the design subsection, this program takes into account more design specifications than those included in the previous design procedure, and it is more elaborate. Once the design is completed, the computer program can be used to analyze the design of the antenna. The description is found in the computer disc available with this book. The program has been developed based on input specifications, which are listed in the program. It can be used as a design tool to determine the geometry of the array (including the number of elements and their corresponding lengths, diameters, and positions) along with the radiation characteristics of the array (including input impedance, VSWR, directivity, front-to-back ratio, E - and H -plane patterns, etc.) based on desired specifications. The input data includes the desired directivity, lower and upper frequency of the operating band, length-to-diameter ratio of the elements, characteristic impedance of the input transmission line, desired input impedance, termination (load) impedance, etc. These and others are listed in the program included in the CD.

The program assumes that the current distribution on each antenna element is sinusoidal. This approximation would be very accurate if the elements were very far from each other. However, in the active region the elements are usually separated by a distance of about 0.1λ when $\alpha = 15^\circ$ and $\tau = 0.9$. Referring to Figure 8.21, one can see that two $\lambda/2$ dipoles separated by 0.1λ have a mutual impedance (almost real) of about 70 ohms. If this mutual impedance is high compared to the resistance of the transmission line (not the characteristic impedance), then the primary method of coupling energy to each antenna will be through the transmission line. If the mutual impedance is high compared to the self-impedance of each element, then the effect on the radiation pattern should be small. In practice, this is usually the case, and the approximation is relatively good. However, an integral equation formulation with a Moment Method numerical solution would be more accurate. The program uses (8-60a) for the self-resistance and (8-60b) for the self-reactance. It uses (8-68) for the mutual impedance, which for the side-by-side configuration reduces to the sine and cosine integrals in [13], similar in form to (8-71a)–(8-71e) for the $l = \lambda/2$ dipole.

The geometry of the designed log-periodic dipole array is that of Figure 11.9, except that the program also allows for an input transmission line (connected to the first/shortest element), a termination transmission line (extending beyond the last/longest element), and a termination (load) impedance. The length of the input transmission line changes the phase of computed data (such as voltage, current, reflection coefficient, etc.) while its characteristic impedance is used to calculate the VSWR, which in turn affects the input impedance measured at the source. The voltages and currents are found based on the admittance method of Kyle [14]. The termination transmission line and the termination (load) impedance allow for the insertion of a matching section whose primary purpose is to absorb any energy which manages to continue past the active region. Without the termination (load) impedance, this energy would be

reflected along the transmission line back into the active region where it would affect the radiation characteristics of the array design and performance.

In designing the array, the user has the choice to select σ and τ (but not the directivity) or to select the directivity (but not σ and τ). In the latter case, the program finds σ and τ by assuming an *optimum design* as defined by the dashed line of Figure 11.13. For the geometry of the array, the program assumes that the elements are placed along the z -axis (with the shortest at $z = 0$ and the longest along the positive z -axis). Each linear element of the array is directed along the y -axis (i.e., the array lies on the yz -plane). The angle θ is measured from the z axis toward the xy -plane while angle ϕ is measured from the x -axis (which is normal to the plane of the array) toward the y -axis along the xy -plane. The E -plane of the array is the yz -plane ($\phi = 90^\circ, 270^\circ; 0^\circ \leq \theta \leq 180^\circ$) while the H -plane is the xz -plane ($\phi = 0^\circ, 180^\circ; 0^\circ \leq \theta \leq 180^\circ$).

11.5 FUNDAMENTAL LIMITS OF ELECTRICALLY SMALL ANTENNAS

In all areas of electrical engineering, especially in electronic devices and computers, the attention has been shifted toward miniaturization. Electromagnetics, and antennas in particular, are of no exception. A large emphasis in the last few years has been placed toward electrically small antennas, including printed board designs. However, there are fundamental limits as to how small the antenna elements can be made. The basic limitations are imposed by the free-space wavelength to which the antenna element must couple to, which has not been or is expected to be miniaturized [15].

An excellent paper on the fundamental limits in antennas has been published [15], and revisited in [16], and most of the material in this section is drawn from [15]. It reviews the limits of electrically small, superdirective, super-resolution, and high-gain antennas. The limits on electrically small antennas are derived by assuming that the entire antenna structure (with a largest linear dimension of $2r$), and its transmission line and oscillator are all enclosed within a sphere of radius r as shown in Figure 11.16(a). Because of the arbitrary current or source distribution of the antenna inside the sphere, its radiated field outside the sphere is expressed as a complete set of orthogonal spherical vector waves or modes. For vertically polarized omnidirectional antennas, only TM_{m0} circularly symmetric (no azimuthal variations) modes are required. Each mode is used to represent a spherical wave which propagates in the outward radial direction. This approach was introduced first by Chu [17], and it was followed by Harrington [18]. Earlier papers on the fundamental limitations and performance of small antennas were published by Wheeler [19], [20]. He derived the limits of a small dipole and a small loop (used as a magnetic dipole) from the limitations of a capacitor and an inductor, respectively. The capacitor and inductor were chosen to occupy, respectively, volumes equal to those of the dipole and the loop.

Using the mathematical formulation introduced by Chu [17], the source or current distribution of the antenna system inside the sphere is not uniquely determined by the field distribution outside the sphere. Since it is possible to determine an infinite number of different source or current distributions inside the sphere, for a given field configuration outside the sphere, Chu [17] confined his interest to the most favorable source distribution and its corresponding antenna structure that could exist within the sphere. This approach was taken to minimize the details and to simplify the task of identifying the antenna structure. It was also assumed that the desired current or source